

1. Here $\mu_1' = E(X_1) = \theta$

(a)

$\hat{\theta} = m_1' = \bar{X}$

(i) $g(\theta) = e^\theta = e^{\mu_1'} = h(\mu_1')$

$\therefore \text{MOME}(e^\theta) = e^{\hat{\theta}_{\text{MOME}}} = e^{m_1'} = e^{\bar{X}} (=h(m_1'))$

$g(\theta) = P_0(X_1+X_2+X_3=1) = 3\theta e^{-3\theta} = 3\mu_1' e^{-3\mu_1'} = h(\mu_1')$, $\hat{g}(\theta) = h(m_1')$

$\therefore \text{MOME}(3\theta e^{-3\theta}) = 3\hat{\theta}_{\text{MOME}} e^{-3\hat{\theta}_{\text{MOME}}} = 3m_1' e^{-3m_1'} = 3\bar{X} e^{-3\bar{X}}$

(b) Here, $\mu_1' = E(X_1) = \alpha\beta$, $\mu_2' = \mu_1'^2 = \alpha\beta^2 \Rightarrow \mu_2' = \alpha(\alpha+1)\beta^2$

Hence, $\alpha = \frac{\mu_1'}{\beta}$, $\alpha+1 = \frac{\mu_1'+\beta}{\beta}$ and $\mu_2' = \mu_1'(\mu_1'+\beta)$,

so that $\beta = \frac{\mu_2'}{\mu_1'} - \mu_1' = \frac{\mu_2' - \mu_1'^2}{\mu_1'}$

and $\alpha = \frac{\mu_1'^2}{\mu_2' - \mu_1'^2} = h_2(\mu_1', \mu_2')$

$\Rightarrow \hat{\beta}_{\text{MOME}} = h_1(m_1', m_2') = \frac{m_2' - m_1'^2}{m_1'} = \frac{\frac{1}{n} \sum (X_i - \bar{X})^2}{\bar{X}} = \frac{(n-1)S^2}{\bar{X}}$

$\alpha_{\text{MOME}} = h_2(m_1', m_2') = \frac{m_1'^2}{m_2' - m_1'^2} = \frac{\bar{X}^2}{(n-1)S^2}$

(c)

$g(\theta) = P(X_1=1) = F_0(1) = 1 - e^{-1/\theta}$

$\therefore \mu_1' = E(X_1) = \theta$, $g(\theta) = 1 - e^{-1/\mu_1'} = h(\mu_1')$

$\therefore \text{MOME}(1 - e^{-1/\theta}) = 1 - e^{-1/\hat{\theta}_{\text{MOME}}} = 1 - e^{-1/m_1'} = 1 - e^{-1/\bar{X}}$

(d)

$\mu_1' = E(X_1) = \theta$

$\mu_2' = V(X_1) + E^2(X_1) = \frac{\theta^2}{12} \Rightarrow \theta = \sqrt{12\mu_2'} = h(\mu_2')$

$\hat{\theta}_{\text{MOME}} = h(m_2')$

$\Rightarrow g(\theta) = \frac{1}{1+\theta}$ has $\text{MOME}\left(\frac{1}{1+\theta}\right) = \frac{1}{1+\hat{\theta}_{\text{MOME}}} = \frac{1}{1+\hat{\theta}_{\text{MOME}}} = \frac{1}{1+\sqrt{12X_{(n)}}}$

2. $\delta_c(X) = \{cX_{(n)} : X_{(n)} = \max_{1 \leq i \leq n} X_i\}$, $g(\theta) = \theta$

$\text{MSE } \delta_c(X) = E_\theta (\delta_c(X) - \theta)^2 = E_\theta (cX_{(n)} - \theta)^2 = \theta^2 \left[\frac{nc^2}{(n+1)^2(n+2)} + \left(\frac{nc}{n+1} - 1 \right)^2 \right]$

Then $\min_c \delta_c(X)$ at $c^* = \frac{n+2}{n+1}$ (Check this!)

$\Rightarrow$ MMSE of $g(\theta)$ in the given class is $\delta_{c^*}(X) = \frac{n+2}{n+1} X_{(n)}$
which has lower MSE than that of the UMVUE
 $\delta(X) = \frac{n+1}{n} X_{(n)}$

likelihood $L(\theta|z) = \prod_{i=1}^{10} P_0(x_i = z_i) = \left(\frac{2\theta}{3}\right)^4 \left(\frac{\theta}{3}\right)^3 \left(\frac{2(1-\theta)}{3}\right)^3 \left(\frac{1-\theta}{3}\right)^3$

log likelihood $l(\theta|z) = \text{const.} + 5 \ln \theta + 5 \ln(1-\theta)$

$$\frac{\partial l(\theta)}{\partial \theta} \Big|_{\hat{\theta}} = 0 \Rightarrow \hat{\theta} = \frac{1}{2}$$

$$\hat{\theta}_{MLE} = \frac{1}{2}$$

check $\frac{\partial^2 l(\theta)}{\partial \theta^2} \Big|_{\hat{\theta}} < 0$

4 a) $L(\theta|z) = \frac{e^{-n\theta} \theta^{\sum x_i}}{\prod x_i!}$, $l(\theta|z) = -n\theta + \sum x_i \ln \theta - \ln(\prod x_i!)$

$$\frac{\partial l(\theta)}{\partial \theta} \Big|_{\hat{\theta}} = 0 \Rightarrow \hat{\theta} = \frac{\sum x_i}{n} = \bar{x}, \quad \frac{\partial^2 l(\theta)}{\partial \theta^2} \Big|_{\hat{\theta}} < 0 \Rightarrow \hat{\theta}_{MLE} = \bar{x}$$

b) ~~$l(\theta|z) = n \ln \theta + \theta \sum \ln x_i - \text{const.}$~~ $l(\theta|z) = n \ln \theta + \theta \sum \ln x_i - \text{const.}$

$$\frac{\partial l(\theta|z)}{\partial \theta} \Big|_{\hat{\theta}} = 0 \Rightarrow \frac{n}{\theta} + \sum \ln x_i = 0 \Rightarrow \hat{\theta} = -\frac{n}{\sum \ln x_i}$$

$$\frac{\partial^2 l(\theta|z)}{\partial \theta^2} \Big|_{\hat{\theta}} < 0 \Rightarrow \hat{\theta}_{MLE} = -\frac{n}{\sum \ln x_i}$$

(Will this MLE always exist?)

c) $L(\theta|z) = \frac{1}{2^n} e^{-\sum_{i=1}^n |x_i - \theta|}$, $l(\theta|z) = \text{const.} - \sum_{i=1}^n |x_i - \theta|$

Maximizing $L(\theta)$ or $l(\theta)$ wrt $\theta \Leftrightarrow$ minimizing $\sum_{i=1}^n |x_i - \theta|$ wrt θ .

$$\Rightarrow \hat{\theta} = \text{median}(x_1, \dots, x_n)$$

and $\hat{\theta}_{MLE} = \text{median}(x_1, \dots, x_n)$ (may not be unique, then?)

d) $L(\theta|z) = \frac{1}{\theta^n} \mathbb{I}(2 \max_i |x_i|, \theta)$

$$= \begin{cases} \frac{1}{\theta^n} & \text{if } \theta > 0, \theta \geq 2 \max_i |x_i| \\ 0 & \text{otherwise} \end{cases}$$

$$L(\theta|z) \text{ max at } \hat{\theta} = 2 \max_i |x_i|$$

$$\hat{\theta}_{MLE} = 2 \max_{1 \leq i \leq n} |x_i|$$

② $L(\underline{\theta}|\underline{x}) = \frac{1}{\theta_2^n} \exp\left(-\frac{1}{\theta_2} \sum_{i=1}^n (x_i - \theta_1)\right) I_{\theta_1, x_{(1)}} \quad , \quad \underline{\theta} = \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} \in \mathbb{R}^2$

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For fixed θ_2 , $L(\underline{\theta})$ max at $\hat{\theta}_1 = x_{(1)} \quad \forall \theta_2$

$\therefore \hat{\theta}_1 = x_{(1)}$
(MLE)

Then $L(\hat{\theta}_1, \theta_2) = \frac{1}{\theta_2^n} \exp\left(-\frac{1}{\theta_2} \sum_{i=1}^n (x_i - x_{(1)})\right)$

$l(\hat{\theta}_1, \theta_2) = -n \ln \theta_2 - \frac{1}{\theta_2} \sum_{i=1}^n (x_i - x_{(1)})$

From $\frac{\partial l}{\partial \theta_2} \Big|_{\hat{\theta}_1, \hat{\theta}_2} = 0 \quad \Delta \quad \frac{\partial^2 l}{\partial \theta_2^2} \Big|_{\hat{\theta}_1, \hat{\theta}_2} < 0$

$\hat{\theta}_2 = \frac{1}{n} \sum_{i=1}^n (x_i - x_{(1)})$

$\Rightarrow \hat{\theta}_2 = \frac{1}{n} \sum_{i=1}^n (x_i - x_{(1)})$
(MLE)

⑥ Done in class, solved completely

⑦ $L(\underline{\theta}|\underline{x}) = \frac{1}{(2\sqrt{3}\theta_2)^n} I_{(\theta_1 - \sqrt{3}\theta_2, x_{(1)})} I_{(x_{(n)}, \theta_1 + \sqrt{3}\theta_2)}$

$\therefore L(\underline{\theta}) > 0$ if $x_{(1)} - \sqrt{3}\theta_2 \leq \theta_1 \leq x_{(1)} + \sqrt{3}\theta_2$, which depends on θ_2

$\therefore$ For given θ_2 , $L(\underline{\theta})$ max wrt θ_1 if $\theta_1 \in [x_{(1)} - \sqrt{3}\theta_2, x_{(1)} + \sqrt{3}\theta_2]$.

One choice would be $\hat{\theta}_1 = \frac{x_{(n)} + x_{(1)}}{2}$ Hence a non-unique MLE of θ_1
(MLE) $\hat{\theta}_1 = \frac{x_{(1)} + x_{(n)}}{2}$

Then $L(\hat{\theta}_1, \theta_2)$ is max wrt θ_2 if $\theta_2 = \frac{x_{(n)} - x_{(1)}}{2\sqrt{3}} (= \hat{\theta}_2)$

$\hat{\theta}_2 = \frac{x_{(n)} - x_{(1)}}{2\sqrt{3}}$
(MLE)

⑧ Here $X_1, \dots, X_n$ i.i.d. discrete uniform distn with pmf

$f_N(x) = \begin{cases} \frac{1}{N} & \text{if } x = 1, 2, \dots, N \\ 0 & \text{otherwise} \end{cases}$

$L(N|\underline{x}) = \begin{cases} \left(\frac{1}{N}\right)^n & \text{if } 1 \leq x_{(n)} \leq N \\ 0 & \text{otherwise} \end{cases}$

max $L(N|\underline{x})$ at $x_{(n)} \Rightarrow \hat{N}_{MLE} = x_{(n)}$

⑨ $f_N(x) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}} = P_N(x)$

$\frac{P_N(x)}{P_{N-1}(x)} = \frac{N-n}{xN} \cdot \frac{N-M}{N-M-n+x} \Rightarrow \frac{P_N(x)}{P_{N-1}(x)} > 1$ iff $N < \frac{nM}{x}$

For a given x (data), $P_N(x)$ attains max at $N = \frac{nM}{x}$

$\therefore \hat{N}_{MLE} = \frac{nM}{x}$

7. $X = \# \text{ sales in a day}$, $X \sim P(\mu)$, $\mu > 0$

Let $Y_i = \begin{cases} 1 & \text{if } 0 \text{ sales in day } i \\ 0 & \text{otherwise} \end{cases}$

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$$P(Y_i = 1) = P(X = 0) = e^{-\mu}$$

Thus, when $X_1, \dots, X_{30}$ are iid $P(\mu)$,
 $Y_1, \dots, Y_{30}$ are iid $\text{Ber}(e^{-\mu})$

$$\text{Then, MLE of } (e^{-\mu}) = e^{-(\text{MLE of } \mu)}$$

$$\text{But MLE of } (e^{-\mu}) = \bar{Y}$$

$$\therefore e^{\hat{\mu}_{\text{MLE}}} = \bar{Y} \Rightarrow \hat{\mu}_{\text{MLE}} = -\ln \bar{Y}$$

$$\text{ML estimate of } \mu \text{ (from the given data)} = -\ln\left(\frac{20}{30}\right)$$

8. Discussed & solved in class

$$\hat{\theta}_{\text{MLE}} = \max(0, \bar{X})$$

9. $X \sim \text{Bin}(1, \frac{1}{1+e^\theta})$, $\theta \in (-\infty, \infty)$

$$L(\theta|1) = \frac{1}{1+e^\theta} \downarrow \theta, \text{ max at } \theta = -\infty, \quad L(\theta|0) = \frac{e^\theta}{1+e^\theta} \uparrow \theta, \text{ max at } \theta = \infty.$$

MLE of θ does not exist.

10. $X_1, \dots, X_n$ r.s. from $P(\theta)$, $\hat{\theta}_{\text{MLE}} = \bar{X}$

$$g(\theta) = \theta^3 (3\sqrt{\theta} + \theta + 12), \text{ MLE of } g(\theta) = \hat{\theta}_{\text{MLE}}^3 (3\sqrt{\hat{\theta}_{\text{MLE}}} + \hat{\theta}_{\text{MLE}} + 12) \\ = (\bar{X}_n)^3 (3\sqrt{\bar{X}_n} + \bar{X}_n + 12),$$

which is a consistent estimator of $g(\theta)$

11. $X_1, \dots, X_n$ r.s. from

$$U(0, \theta), \quad g(\theta) = e^\theta$$

$$\hat{\theta}_{\text{MLE}} = X_{(n)}, \text{ MLE of } (e^\theta) = e^{\hat{\theta}_{\text{MLE}}} = e^{X_{(n)}}, \text{ consistent estimator of } e^\theta.$$

12. $X_1, \dots, X_n$ r.s. from $P(\theta)$, $\hat{\theta}_{\text{MLE}} = \bar{X}_n$, $\sqrt{n}(\bar{X}_n - \theta) \xrightarrow{L} N(0, \frac{1}{I(\theta)})$
 i.e., $\sqrt{n}(\bar{X}_n - \theta) \rightarrow N(0, \theta)$

$$g(\bar{X}_n) = e^{\bar{X}_n} \text{ is MLE of } g(\theta) = e^\theta, \text{ by } \Delta\text{-rule} \\ \sqrt{n}(e^{\bar{X}_n} - e^\theta) \xrightarrow{L} N(0, \theta e^{-2\theta})$$

13. $X_1, \dots, X_n$ r.s. from $N(0, 1/\theta)$

$$\hat{\theta}_{\text{MLE}} = \frac{n}{\sum X_i^2}$$

$$\sqrt{n}\left(\frac{1}{\hat{\theta}_{\text{MLE}}} - \frac{1}{\theta}\right) \xrightarrow{L} N\left(0, \frac{(g'(\theta))^2}{I(\theta)}\right) \text{ with } I(\theta) = \frac{1}{2\theta^2} \\ \text{i.e., } \sqrt{n}\left(\left(\frac{1}{\theta}\right)_{\text{MLE}} - \frac{1}{\theta}\right) \xrightarrow{L} N\left(0, \frac{2}{\theta^2}\right)$$